

Research Notes on Long short-term memory

Long short-term memory

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

$$h_t = o_t \odot \tanh(c_t)$$

>> Section 1

CTC score function Many applications use stacks of LSTM RNNs and train them by connectionist temporal classification (CTC) to find an RNN weight matrix that maximizes the probability of the label sequences in a training set, given the corresponding input sequences. CTC achieves both alignment and recognition.

>> Section 2

$W \in \mathbb{R}^{h \times d}$, $U \in \mathbb{R}^{h \times h}$ and $b \in \mathbb{R}^h$: weight matrices and bias vector parameters which need to be learned during training

>> Section 3

Further reading Monner, Derek D.; Reggia, James A. (2010). "A generalized LSTM-like training algorithm for second-order recurrent neural networks" (PDF). *Neural Networks*. 25 (1): 70–83. doi:10.1016/j.neunet.2011.07.003. PMC 3217173. PMID 21803542. High-performing extension of LSTM that has been simplified to a single node type and can train arbitrary architectures Gers, Felix A.; Schraudolph, Nicol N.; Schmidhuber, Jürgen (Aug 2002). "Learning precise timing with LSTM recurrent networks" (PDF). *Journal of Machine Learning Research*. 3: 115–143. Gers, Felix (2001). "Long Short-Term Memory in Recurrent Neural Networks" (PDF). PhD thesis. Abidogun, Olusola Adeniyi (2005). *Data Mining, Fraud Detection and Mobile Telecommunications: Call Pattern Analysis with Unsupervised Neural Networks*. Master's Thesis (Thesis). University of the Western Cape. hdl:11394/249. Archived (PDF) from the original on May 22, 2012. original with two chapters devoted to explaining recurrent neural networks, especially LSTM.

>> Section 4

The figure on the right is a graphical representation of an LSTM unit with peephole connections (i.e. a peephole LSTM). Peephole connections allow the gates to access the constant error carousel (CEC), whose activation is the cell state. h_{t-1} is not used, c_{t-1} is used instead in most places.

>> Section 5

Long short-term memory (LSTM) is a type of recurrent neural network (RNN) aimed at mitigating the vanishing gradient problem commonly encountered by traditional RNNs. Its relative insensitivity to gap length is its advantage over other RNNs, hidden Markov models, and other sequence learning methods. It aims to provide a short-term memory for RNN that can last thousands of timesteps (thus "long short-term memory"). The name is made in analogy with long-term memory and short-term memory and their relationship, studied by cognitive psychologists since the early 20th century. An LSTM unit is typically composed of a cell and three gates: an input gate, an output gate, and a forget gate. The cell remembers values over arbitrary time intervals, and the gates regulate the flow of information into and out of the cell. Forget gates decide what information to discard from the previous state, by mapping the previous state and the current input to a value between 0 and 1. A (rounded) value of 1 signifies retention of the information, and a value of 0 represents discarding. Input gates decide which pieces of new information to store in the current cell state, using the same system as forget gates. Output gates control which pieces of information in the current cell state to output, by assigning a value from 0 to 1 to the information, considering the previous and current states. Selectively outputting relevant information from the current state allows the LSTM network to maintain useful, long-term dependencies to make predictions, both in current and future time-steps. LSTM has wide applications in classification, data processing, time series analysis tasks, speech recognition, machine translation, speech activity detection, robot control, video games, healthcare, energy forecasting.

>> Section 6

Training An RNN using LSTM units can be trained in a supervised fashion on a set of training sequences, using an optimization algorithm like gradient descent combined with backpropagation through time to compute the gradients needed during the optimization process, in order to change each weight of the LSTM network in proportion to the derivative of the error (at the output layer of the LSTM network) with respect to corresponding weight. A problem with using gradient descent for standard RNNs is that error gradients vanish exponentially quickly with the size of the time lag between important events. This is due to $\lim_{n \rightarrow \infty} W^n = 0$ if the spectral radius of W is smaller than 1. However, with LSTM units, when error values are back-propagated from the output layer, the error remains in the LSTM unit's cell. This "error carousel" continuously feeds error back to each of the LSTM unit's

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gates, until they learn to cut off the value.

>> Section 7

Motivation In theory, classic RNNs can keep track of arbitrary long-term dependencies in the input sequences. The problem with classic RNNs is computational (or practical) in nature: when training a classic RNN using back-propagation, the long-term gradients which are back-propagated can "vanish", meaning they can tend to zero due to very small numbers creeping into the computations, causing the model to effectively stop learning. RNNs using LSTM units partially solve the vanishing gradient problem, because LSTM units allow gradients to also flow with little to no attenuation. However, LSTM networks can still suffer from the exploding gradient problem. The intuition behind the LSTM architecture is to create an additional module in a neural network that learns when to remember and when to forget pertinent information. In other words, the network effectively learns which information might be needed later on in a sequence and when that information is no longer needed. For instance, in the context of natural language processing, the network can learn grammatical dependencies. An LSTM might process the sentence "Dave, as a result of his controversial claims, is now a pariah" by remembering the (statistically likely) grammatical gender and number of the subject Dave. Note that this information is pertinent for the pronoun his and note that this information is no longer important after the verb is.

>> Section 8

Development Aspects of LSTM were anticipated by "focused back-propagation", cited by the LSTM paper. Sepp Hochreiter's 1991 German diploma thesis analyzed the vanishing gradient problem and developed principles of the method. His supervisor, Jürgen Schmidhuber, considered the thesis highly significant. An early version of LSTM was published in 1995 in a technical report by Sepp Hochreiter and Jürgen Schmidhuber, then published in the NIPS 1996 conference. The most commonly used reference point for LSTM was published in 1997 in the journal Neural Computation. By introducing Constant Error Carousel (CEC) units, LSTM deals with the vanishing gradient problem. The initial version of LSTM block included cells, input and output gates. Felix Gers, Jürgen Schmidhuber, and Fred Cummins introduced the forget gate (also called "keep gate") into the LSTM architecture in 1999, enabling the LSTM to reset its own state. This is the most commonly used version of LSTM nowadays. They added peephole connections in 2000. Additionally, the output activation function was omitted.